University of Wah

Department of Computer Science

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Course:

Computer Network_Lab - CS-241L

Reg No:

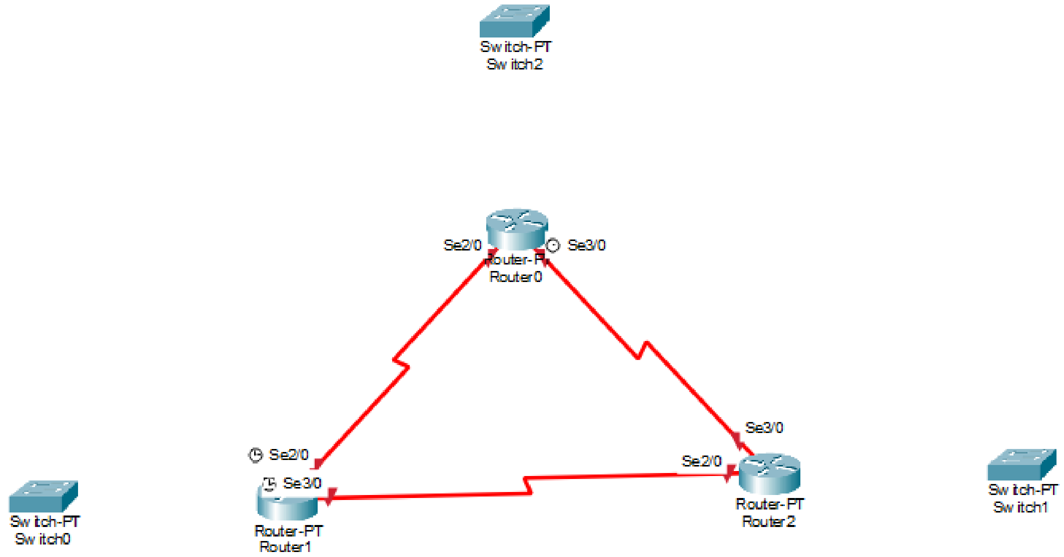
UW-24-CS-BS-101

Section:

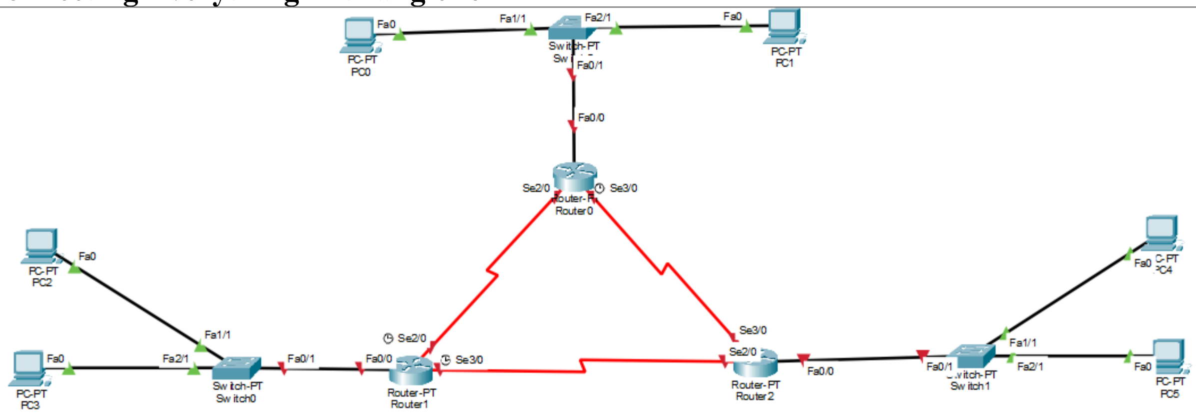
BS-CS-3rd-B

Question # 01:

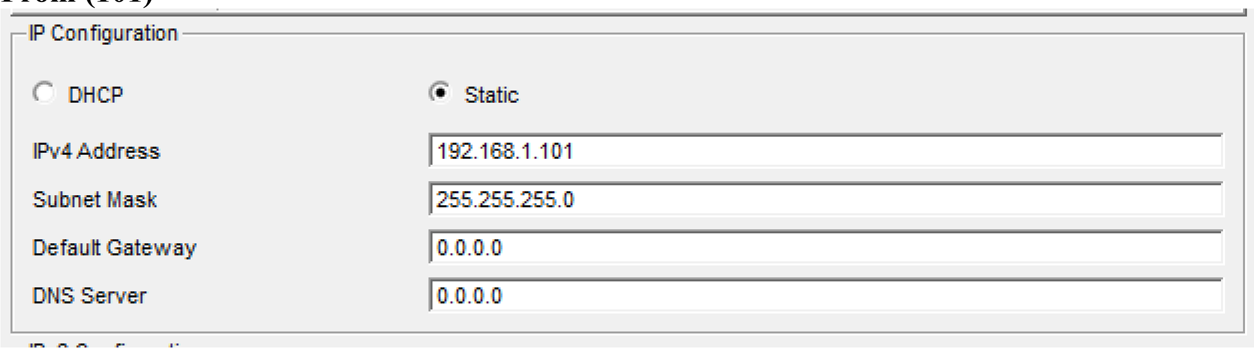
Adding and connecting all PC , router, switches



Connecting Everything in triangle form



Giving IP address to PC
From (101)


IP Configuration

DHCP   Static

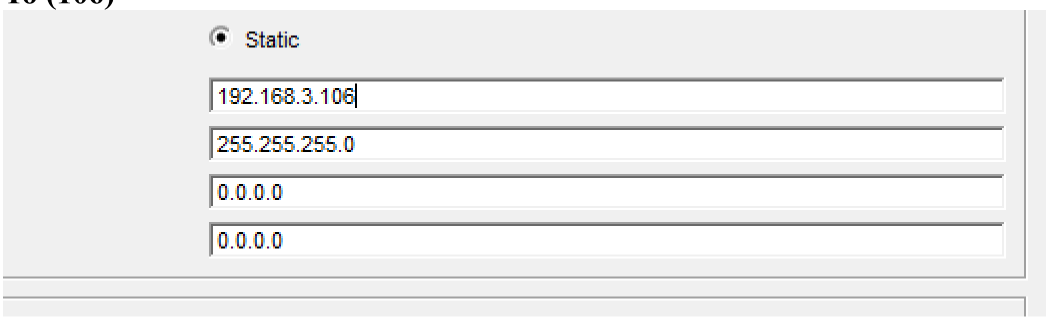
IPv4 Address: 192.168.1.101

Subnet Mask: 255.255.255.0

Default Gateway: 0.0.0.0

DNS Server: 0.0.0.0

To (106)


Static

192.168.3.106

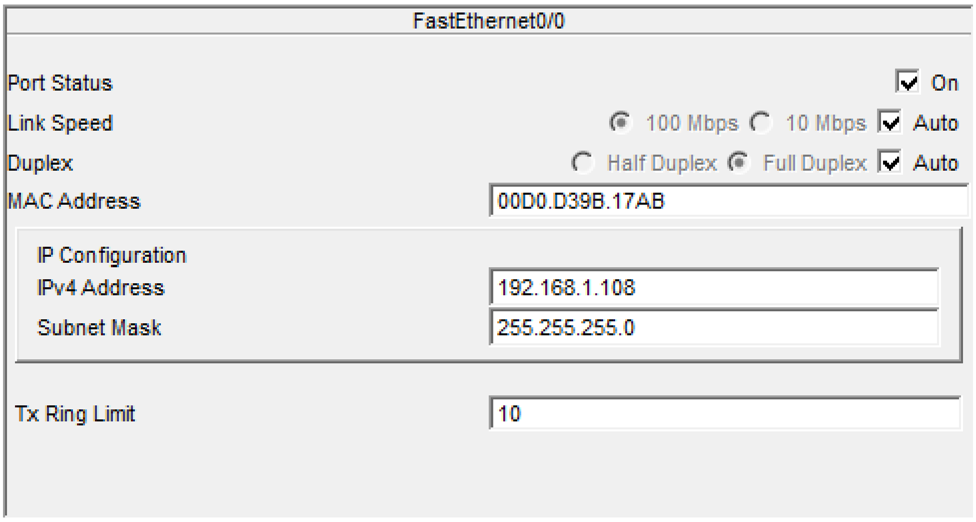
255.255.255.0

0.0.0.0

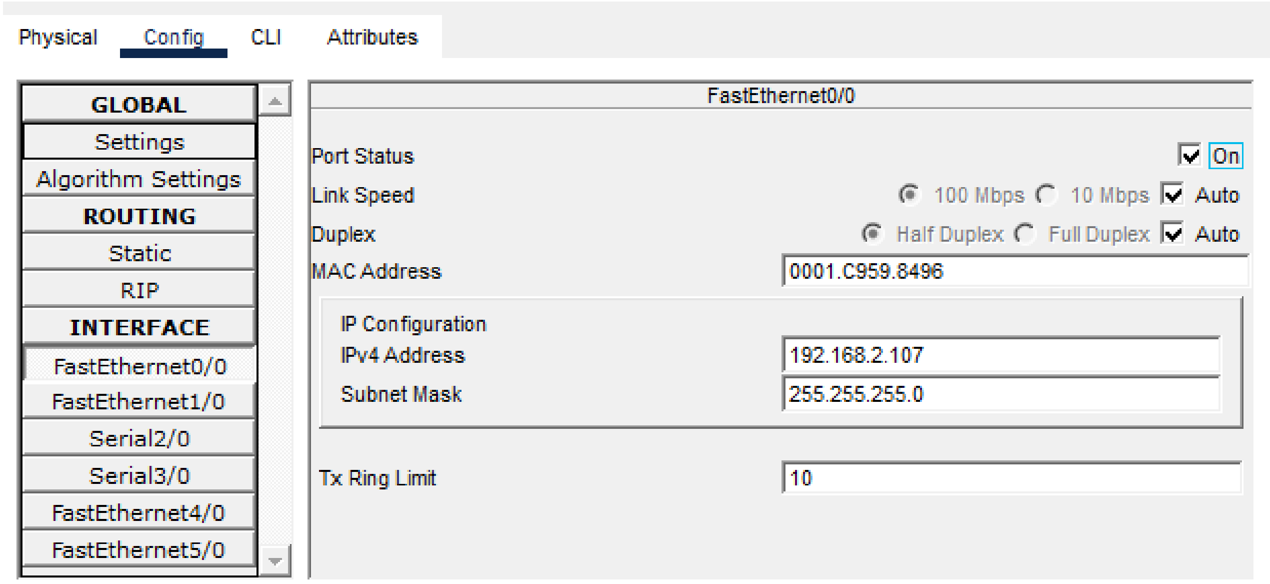
0.0.0.0

Giving Router IP address

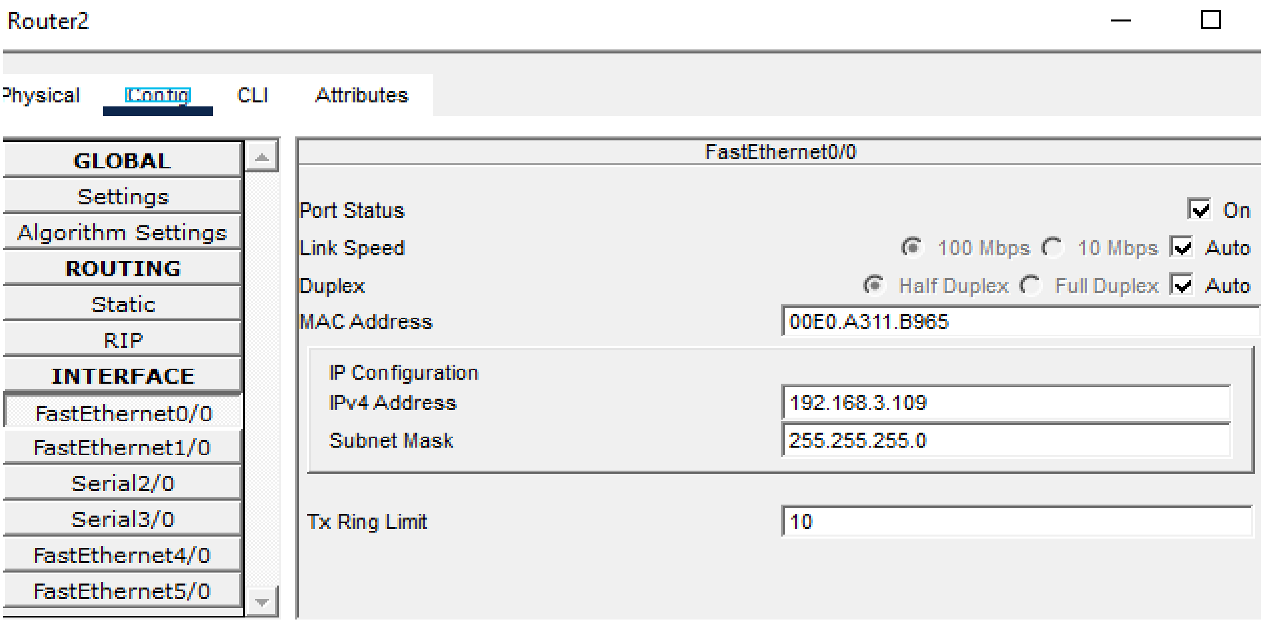
Router 1



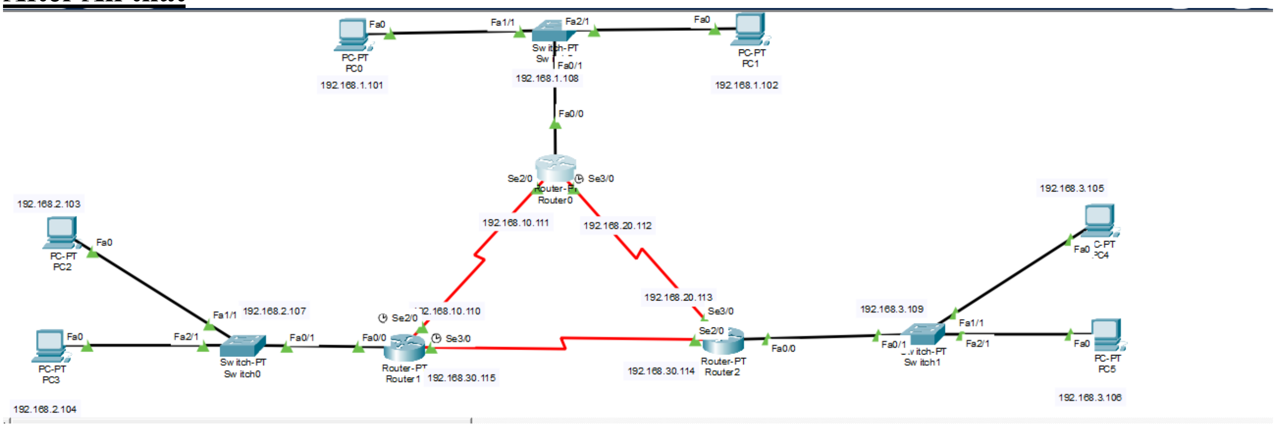
Router 2



Router 3



After All that



Applying Static Routing from static Option

Router 1:

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#ip route 192.168.2.0 255.255.255.0 192.168.10.110
Router(config)#ip route 192.168.3.0 255.255.255.0 192.168.20.113
Router(config)#
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Router 2:

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#ip route 192.168.1.0 255.255.255.0 192.168.10.111
Router(config)#ip route 192.168.3.0 255.255.255.0 192.168.30.114
Router(config)#
Router(config)#end
Router#
```

Router 3:

```
Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#ip route 192.168.1.0 255.255.255.0 192.168.20.112
Router(config)#ip route 192.168.2.0 255.255.255.0 192.168.30.115
Router(config)#
Router(config)#end
Router#
```

Sending Messages throught ping

From PC1(Router 1)

To PC3(Router 2)

```
C:\>ping 192.168.2.103

Pinging 192.168.2.103 with 32 bytes of data:

Reply from 192.168.2.103: bytes=32 time=1ms TTL=126
Reply from 192.168.2.103: bytes=32 time=1ms TTL=126
Reply from 192.168.2.103: bytes=32 time=1ms TTL=126
Reply from 192.168.2.103: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.2.103:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 1ms, Average = 1ms
```

From PC1(Router 1)
To PC5(Router 3)

```
C:\>ping 192.168.3.105

Pinging 192.168.3.105 with 32 bytes of data:

Request timed out.
Reply from 192.168.3.105: bytes=32 time=17ms TTL=125
Reply from 192.168.3.105: bytes=32 time=1ms TTL=125
Reply from 192.168.3.105: bytes=32 time=20ms TTL=125

Ping statistics for 192.168.3.105:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 20ms, Average = 12ms
```

From PC2(Router 2)
To PC1(Router 1)

```
C:\>ping 192.168.1.101

Pinging 192.168.1.101 with 32 bytes of data:

Reply from 192.168.1.101: bytes=32 time=1ms TTL=126
Reply from 192.168.1.101: bytes=32 time=1ms TTL=126
Reply from 192.168.1.101: bytes=32 time=1ms TTL=126
Reply from 192.168.1.101: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.1.101:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 1ms, Average = 1ms
```

From PC1(Router 2)
To PC5(Router 3)

```
C:\>ping 192.168.3.106

Pinging 192.168.3.106 with 32 bytes of data:

Request timed out.
Reply from 192.168.3.106: bytes=32 time=13ms TTL=125
Reply from 192.168.3.106: bytes=32 time=10ms TTL=125
Reply from 192.168.3.106: bytes=32 time=12ms TTL=125

Ping statistics for 192.168.3.106:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 10ms, Maximum = 13ms, Average = 11ms
```

From PC2(Router 3)
To PC1(Router 1)

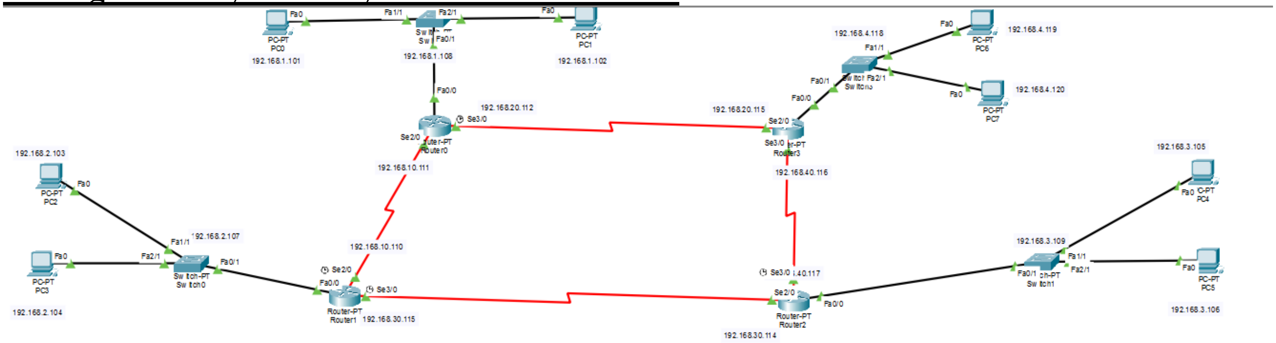
```
C:\>ping 192.168.2.104

Pinging 192.168.2.104 with 32 bytes of data:

Reply from 192.168.2.104: bytes=32 time=1ms TTL=126
Reply from 192.168.2.104: bytes=32 time=1ms TTL=126
Reply from 192.168.2.104: bytes=32 time=1ms TTL=126
Reply from 192.168.2.104: bytes=32 time=11ms TTL=126

Ping statistics for 192.168.2.104:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 11ms, Average = 3ms
```

From PC1(Router 3)
To PC5(Router 2)

```
C:\>ping 192.168.3.106

Pinging 192.168.3.106 with 32 bytes of data:

Request timed out.
Reply from 192.168.3.106: bytes=32 time=13ms TTL=125
Reply from 192.168.3.106: bytes=32 time=10ms TTL=125
Reply from 192.168.3.106: bytes=32 time=12ms TTL=125

Ping statistics for 192.168.3.106:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 10ms, Maximum = 13ms, Average = 11ms
```

Question # 03

Adding 1 Router, 1 Switch, and 2 PCs in Network



Applying RIP to All Router

Router 1

ROUTING
Static
RIP
INTERFACE
FastEthernet0/0
FastEthernet1/0
Serial2/0
Serial3/0

Network Address
192.168.1.0
192.168.10.0
192.168.20.0

Router 2

ROUTING
Static
RIP
INTERFACE
FastEthernet0/0
FastEthernet1/0
Serial2/0

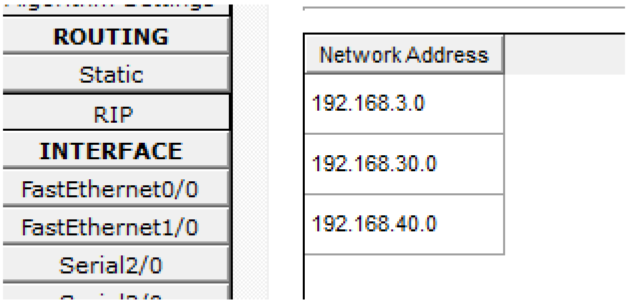
Network Address
192.168.2.0
192.168.10.0
192.168.30.0

Router 3

ROUTING
Static
RIP
INTERFACE
FastEthernet0/0
FastEthernet1/0
Serial2/0

Network Address
192.168.4.0
192.168.20.0
192.168.40.0

Router 4



Sending Messages Through to Ping

From PC1(Router 4)
To PC5(Router 3)

```
C:\>ping 192.168.4.119

Pinging 192.168.4.119 with 32 bytes of data:

Reply from 192.168.4.119: bytes=32 time=3ms TTL=124
Reply from 192.168.4.119: bytes=32 time=3ms TTL=124
Reply from 192.168.4.119: bytes=32 time=10ms TTL=124
Reply from 192.168.4.119: bytes=32 time=3ms TTL=124

Ping statistics for 192.168.4.119:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 10ms, Average = 4ms
```

From PC1(Router 3)
To PC2(Router 1)

```
C:\>ping 192.168.1.101

Pinging 192.168.1.101 with 32 bytes of data:

Reply from 192.168.1.101: bytes=32 time=3ms TTL=124
Reply from 192.168.1.101: bytes=32 time=4ms TTL=124
Reply from 192.168.1.101: bytes=32 time=3ms TTL=124
Reply from 192.168.1.101: bytes=32 time=3ms TTL=124

Ping statistics for 192.168.1.101:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 4ms, Average = 3ms
```

From PC2(Router 2)
To PC1(Router 4)

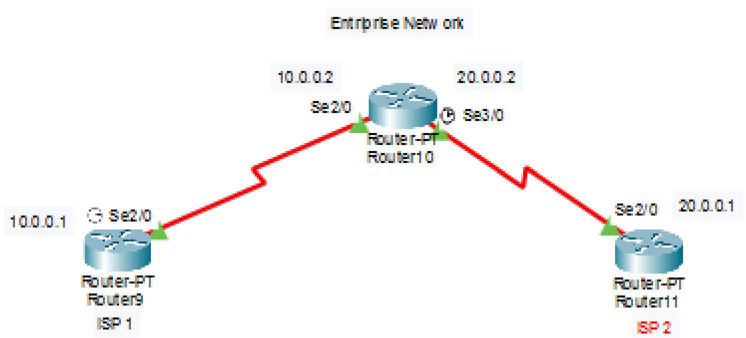
```
C:\>ping 192.168.2.104

Pinging 192.168.2.104 with 32 bytes of data:

Reply from 192.168.2.104: bytes=32 time=1ms TTL=126
Reply from 192.168.2.104: bytes=32 time=1ms TTL=126
Reply from 192.168.2.104: bytes=32 time=1ms TTL=126
Reply from 192.168.2.104: bytes=32 time=11ms TTL=126

Ping statistics for 192.168.2.104:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 11ms, Average = 3ms
```

From PC1(Router 1)
To PC5(Router 4)

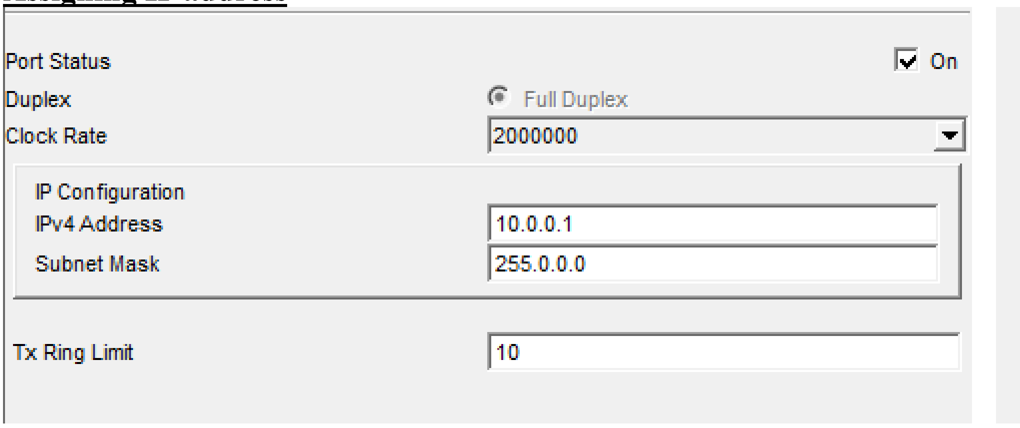
```
C:\>ping 192.168.3.106

Pinging 192.168.3.106 with 32 bytes of data:

Request timed out.
Reply from 192.168.3.106: bytes=32 time=13ms TTL=125
Reply from 192.168.3.106: bytes=32 time=10ms TTL=125
Reply from 192.168.3.106: bytes=32 time=12ms TTL=125

Ping statistics for 192.168.3.106:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 10ms, Maximum = 13ms, Average = 11ms
```

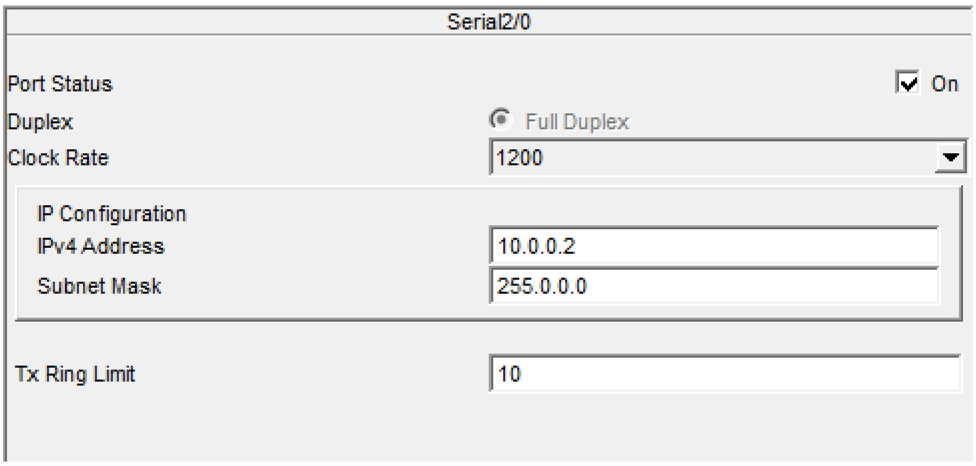
Question # 02

Network



Assigning IP address


Port Status On
Duplex Full Duplex
Clock Rate 2000000
IP Configuration
IPv4 Address 10.0.0.1
Subnet Mask 255.0.0.0
Tx Ring Limit 10


Serial2/0
Port Status On
Duplex Full Duplex
Clock Rate 1200
IP Configuration
IPv4 Address 10.0.0.2
Subnet Mask 255.0.0.0
Tx Ring Limit 10

Port Status	☑ On
Duplex	Full Duplex
Clock Rate	2000000
IP Configuration	
IPv4 Address	20.0.0.2
Subnet Mask	255.0.0.0
Tx Ring Limit	10

Serial2/0

Port Status	☑ On
Duplex	Full Duplex
Clock Rate	1200
IP Configuration	
IPv4 Address	20.0.0.1
Subnet Mask	255.0.0.0
Tx Ring Limit	10

Giving BGP 300 to Entriprise:

```
Router(config-if)#router bgp 300
Router(config-router)#neighbor 10.0.0.1 remote-as 100
Router(config-router)#neighbor 20.0.0.1 remote-as 200
Router(config-router)#network 192.168.40.0 mask 255.255.255.0
Router(config-router)#
```

Giving BGP 100 to ISP1:

```
Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router bgp 100
Router(config-router)#neighbor 10.0.0.2 remote-as 300
Router(config-router)#%BGP-5-ADJCHANGE: neighbor 10.0.0.2 Up

Router(config-router)#
```

Giving BGP 200 to ISP2:

```
Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router bgp 200
Router(config-router)#neighbor 20.0.0.2 remote-as 300
Router(config-router)#%BGP-5-ADJCHANGE: neighbor 20.0.0.2 Up

Router(config-router)#
```

Summary Through Command:

```
Router>show ip bgp summary
BGP router identifier 20.0.0.2, local AS number 300
BGP table version is 1, main routing table version 6
0 network entries using 0 bytes of memory
0 path entries using 0 bytes of memory
0/0 BGP path/bestpath attribute entries using 0 bytes of memory
0 BGP AS-PATH entries using 0 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory
BGP using 32 total bytes of memory
BGP activity 0/0 prefixes, 0/0 paths, scan interval 60 secs

Neighbor        V    AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
10.0.0.1        4   100       4       4        1    0    0 00:02:23        4
20.0.0.1        4   200       3       3        1    0    0 00:01:26        4

Router>show ip bgp
BGP table version is 1, local router ID is 20.0.0.2
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop            Metric LocPrf Weight Path

Router>
```

Testing:

Successful	Route...	Router9	ICMP		0.000	N	0	(e...	(delete)
Successful	Route...	Router11	ICMP		0.000	N	1	(e...	(delete)